

Original Article

Modified Xiaoyaosan reverses aberrant brain regional homogeneity to exert antidepressive effects in mice

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The aim of this study is to explore the macroscale neural mechanisms of the antidepressant effects of Xiaoyaosan, a traditional Chinese herbal formula. We analyzed blood oxygen level-dependent (BOLD) functional magnetic resonance imaging signals. To further minimize the hemodynamic variations of BOLD signals and analyze the intra-region neural activity or temporal coherence, we employed the newly developed regional homogeneity (ReHo) approach to determine aberrant functional connectivity using the chronic unpredictable mild stress (CUMS) mouse model of depression and to also explore the brain-region rescue effect of modified Xiaoyaosan (MXYS) in such mice. We found the aberrant ReHo in CUMS mice replicated previous discoveries in patients with depression. Intriguingly, MXYS only normalized several limbic regions, which suggests the essential roles of these regions in mediating the antidepressant effects of MXYS. Our results provide a reliable framework for the use of ReHo analysis with animal models of depression and further suggest a new perspective to elucidate the antidepressant effects of MXYS.

Key words: depression, fMRI, hippocampus, MXYS, regional homogeneity.

INTRODUCTION

Major depressive disorder is the most common neuropsychiatric disease, characterized by the persistence of

negative mood and loss of pleasure. It has surpassed heart diseases and cancer to become the leading cause of disability worldwide,^{1,2} affecting more than 320 million people, with a rapid increase of prevalence between 2005 and 2015.³ The current severe status of depression urgently demands more efficient antidepressant drugs, as traditional antidepressants are associated with delayed therapeutic responses, along with low responsive and remissive rates among patients with depression.^{4,5}

Accumulated evidence has shown that Xiaoyaosan, a traditional Chinese herbal formula, exerts antidepressant effects via improving the core symptoms of behavioral despair and anhedonia.^{6–9} Xiaoyaosan has been prescribed to treat mental disorders for thousands of years in China^{6,10} and was originally recorded in *Taiping Huimin Heji Jufang*, the first official compilation of materia medica around the world during the Song Dynasty in China (960–1127 AD). In our previous studies, we discovered that Xiaoyaosan and its modified formulas might ameliorate hypothalamic–pituitary–adrenal hyperactivity, reverse neurotrophic abnormalities, and maintain the internal homeostasis of the kynurenine/serotonin pathway.^{8–10} Despite the progress in the study of the molecular mechanisms related to Xiaoyaosan's ability to alleviate depression-like behaviors, the evidence on the implicated macroscale neural mechanisms is far from sufficient.

To determine macroscale neural activities, various functional imaging techniques in humans have revealed that patients with depression exhibit widespread abnormal functional connectomics throughout the brain,¹¹ which is partially reversed by antidepressant therapies.¹² Using functional magnetic resonance imaging (fMRI), we previously demonstrated using amplitude of low-frequency fluctuations (ALFFs) analysis^{13,14} that blood oxygen level-dependent (BOLD) signals were significantly upregulated, primarily in the limbic system, in the chronic unpredictable mild stress (CUMS) mouse model of depression. Intriguingly, we

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Received 29 December 2018; Revised 22 January 2019; Accepted 24 January 2019; published online 04 March 2019.

Table 1 The putative major ingredients and targets of 10 herbs in MXYS

Latin name	English name	Chinese name	Dose (g)	Number of screening ingredients	Major ingredients
<i>Radix bupleuri</i>	Chinese thorowax root	柴胡	15	6	Aikosaponin a,b,c,d Rutin Ribitol A-spinach sterol Volatile oil Polysaccharide
<i>Radix angelicae sinensis</i>	Chinese angelica	当归	15	19	Ligustilide Ferulic acid Camphoric acid Azelaic acid Myrcene Angelica Xanthine Vanillin P-cresol Sphingomyelin Phosphatidylinositol Vanillic acid New angelica lactone Succinic acid Nicotinic acid Uracil Vitamin B ₁₂ Vitamin A β-glutamol Paeoniflorin Paeoniflorin oxide Benzoic acid β-glutamol Tannin and volatile oil
<i>Radices paeoniae alba</i>	Debark peony root	白芍	15		α-humu-lene β-humu-lene β-elemol α-curcumene α-tractlone 3β-acetoxyatractylone Selina-4,7-diene-8-one Eudesmol Palmitic acid Hinesol β-selinene Atractylenolide-I,-II,-III 8β-ethoxyatractylenolide-II 14-acetyl-12-senecieryl-2E,8Z,10E-atracetylenetriol 14-acetyl-12-senecieryl-2E,8E,10E-atractylenetriol 12-senecieryl-2E,8Z,10E-atracetylenetriol 12-senecieryl-2E,8E,10E-atractylenetriol 12α-methyl butyryl-14-acetyl-2E,8Z,10E-atractylenetriol 12α-methylbutyryl-14-acetyl-2E,8E,10E-atractylenetriol 14α-methyl butyryl-2E,8Z,10E-atractylenetriol 14α-methyl butyryl-2E,8E,10E-atractylenetriol Scopoletin Fructose Inulin Mannan AM-3 Aspartic acid Serine Glutamic acid Alanine
<i>Rhizoma atracylodes macrocephalae</i>	Large-head atractylodes rhizome	白术	15	39	

Table 1 (Continued)

Latin name	English name	Chinese name	Dose (g)	Number of screening ingredients	Major ingredients
<i>Rhizoma acori tatarinowii</i>	Grassleaf sweetflag rhizome	石菖蒲	12	5	Glycine Valine Isoleucine Leucine Tyro-sine Phenylalanine Lysine Histisine Arginine Proline α -, β -, γ -asarone Asarylaldehyde Bisasarine Eugenol <i>Cinnamomum camphora</i>
<i>Curcuma aromatica</i>	Turmeric root tuber	郁金	15	3	beta-Sitosterol Sitosterol Naringenin
<i>Caulis polygoni multiflori</i>	Tuber fleeceflower stem	夜交藤/首乌藤	30	3	Emodin Chrysophanol Emodin methyl ether
<i>Fructus schisandrae chinensis</i>	Chinese magnoliavine fruit	五味子	10	11	Fruit: Volatile oil Organic acid Vitamin Seed: Fatty oil Deoxyschizandrin Gamma schisandrin Propane Schisandrin Gomisin A Wu-weiziester A GomisinB
<i>Semen ziziphi spinosae</i>	Spine date seed	酸枣仁	20	5	Zizyphus Zizyphi base Alumina Zizyphus Zizyphi annulus peptide <i>Betula platyphylla</i> Carotene
<i>Os draconis</i>	Bone fossil of big mammals	龙骨	30	7	Calcium carbonate calcium sulfate Iron Potassium Sodium Chloride Sulfate

MXYS, modified Xiaoyaosan.

further uncovered that modified Xiaoyaosan (MXYS) normalized BOLD signals to exert antidepressant effects in CUMS mice.⁶ However, BOLD analysis involves technical challenges because the hemodynamic response may vary among regions¹⁵ and even among trials.^{16,17} In comparison, a recently developed regional homogeneity (ReHo) approach¹⁸ is able to minimize the hemodynamic variations and analyze the intra-region neural activity or temporal coherence. In this study, we employed CUMS as a reliable

mouse model to induce depression and further applied this ReHo analysis to accurately determine the neural activities of dysfunctional brain regions in the resting state.

MATERIALS AND METHODS

Animals

All the procedures used in the animal studies were approved by the National Institutional Animal Care and

Ethical Committee of Southern Medical University. All the experimental methods performed in this study were in accordance with the approved guidelines. Eight-week-old male C57BL/6J mice (Animal Experimental Center, Southern Medical University, China) were randomly divided into three groups ($N = 10$): (i) control group; (ii) CUMS group; and (iii) CUMS + MXYS treated group. All animals were housed in an animal facility where temperature and humidity are strictly controlled ($21 \pm 2^\circ\text{C}$ and $55\% \pm 5\%$) with a light-dark cycle of 12:12 h.

CUMS procedure

The CUMS procedure included nine different stressors, which were randomly arranged throughout 42 consecutive days. The stressors were: (i) 24-h fasting and water deprivation; (ii) 1-h exposure to an empty bottle; (iii) 17-h exposure to a 45° tilting cage; (iv) overnight illumination; (v) 24-h exposure to a wet cage; (vi) swimming in cold water (4°C) for 5 min; (vii) 24-h disruption of the cage; (viii) 24-h foreign body stimulation; and (ix) 4-h restraint in a tube.

Preparation of MXYS

As we have previously described,⁶ MXYS consists of 10 herbs (Table 1). In accordance with the Regulation on Processing of Traditional Chinese Medical Herbal Pieces, all raw herbs were purchased from Nanfang Hospital and processed into a no-decocted extract by EFONG Pharmaceutical Company (Guangdong, China). The putative major ingredients, targets of 10 herbs in MXYS, and the production procedure of no-decocted MXYS have been described in our previous studies.⁶ The quality of MXYS was controlled by regular items regarding granules under the 2010 edition of the *Pharmacopeia*. The drug was packaged into a medicinal composite film, in line with the pharmaceutical packaging specifications. In order to confirm the quality control, critical compounds from MXYS samples were previously identified by employing an high-performance liquid chromatography LTQ-Orbitrap mass spectrometry eluted system.⁶

Drug treatment

As previously described,⁶ all drug doses were converted according to the human and animal equivalent dose

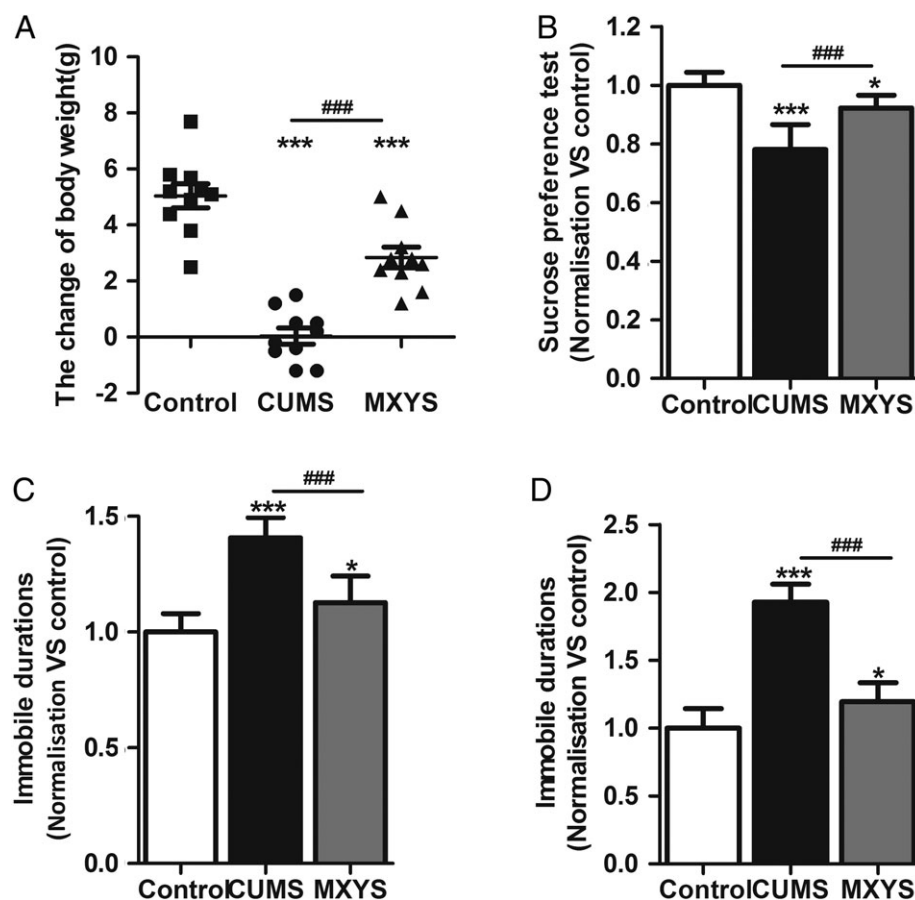


Fig. 1 Effects of MXYS on the body weight change and behavior of mice subjected to CUMS. (A) The difference of body weight gain at the endpoint of the animal treatment. (B) Sucrose preference. (C) TST immobile time. (D) FST immobile time. Data are expressed as the mean + SD. * $P < 0.05$, *** $P < 0.001$ versus control group; ### $P < 0.001$ versus CUMS group. CUMS, chronic unpredictable mild stress; FST, forced swimming test; MXYS, modified Xiaoyaosan; TST, tail suspension test.

Table 2 Quantitative analysis of ReHo between CUMS and control groups

Group	Increased structural name	T-average	Coordinates of the peak voxel			Decreased structural name	T-average	Coordinates of the peak voxel		
			X	Y	Z			X	Y	Z
CUMS vs. control	Infralimbic cortex	6.2053	0.34168	4.1659	3.4821					
	Subiculum	10.796	4.9004	4.3571	-7.3179	Amygdala	5.1459	-5.2862	8.5432	-5.1579
	Ventral orbital cortex	4.6102	1.815	5.1483	3.2421	Cerebellum: flocculonodular lobe	5.8402	-6.413	5.751	-10.6779
	Caudate putamen	9.3871	-3.8854	3.8001	-0.1179	Insular cortex	4.5212	5.3359	7.3997	-0.1179
	Prelimbic cortex	5.6592	0.34498	4.0412	3.2421	Sensory cortex	12.0614	4.4945	3.4471	2.7621
	Cingulate cortex	8.3302	-0.30343	1.5019	1.8021	Auditory cortex	6.5125	-6.616	3.5437	-5.6379
	Formix	4.3817	-0.40738	5.6285	-0.3579					
	Hippocampus	10.796	4.9004	4.3571	-7.3179					
	Midbrain: inferior colliculus	9.6403	0.78056	3.5366	-8.9979					
	Midbrain: periaqueductal gray matter	6.8894	-1.1069	5.0743	-7.7979					
	Midbrain: superior colliculus	9.1174	0.22289	3.3566	-7.3179					
	Motor cortex	8.7917	-2.5919	1.4101	3.0021					
	Olfactory cortex	4.0441	6.3242	6.0422	-3.9579					
	Orbital cortex	6.0284	1.815	5.0023	3.2421					
	Parietal association cortex	4.6599	2.0411	0.9336	-3.4779					

The voxel-level height threshold was $P < 0.005$ (uncorrected) and the cluster-extent threshold was 20 voxels. CUMS, chronic unpredictable mild stress; ReHo, regional homogeneity.

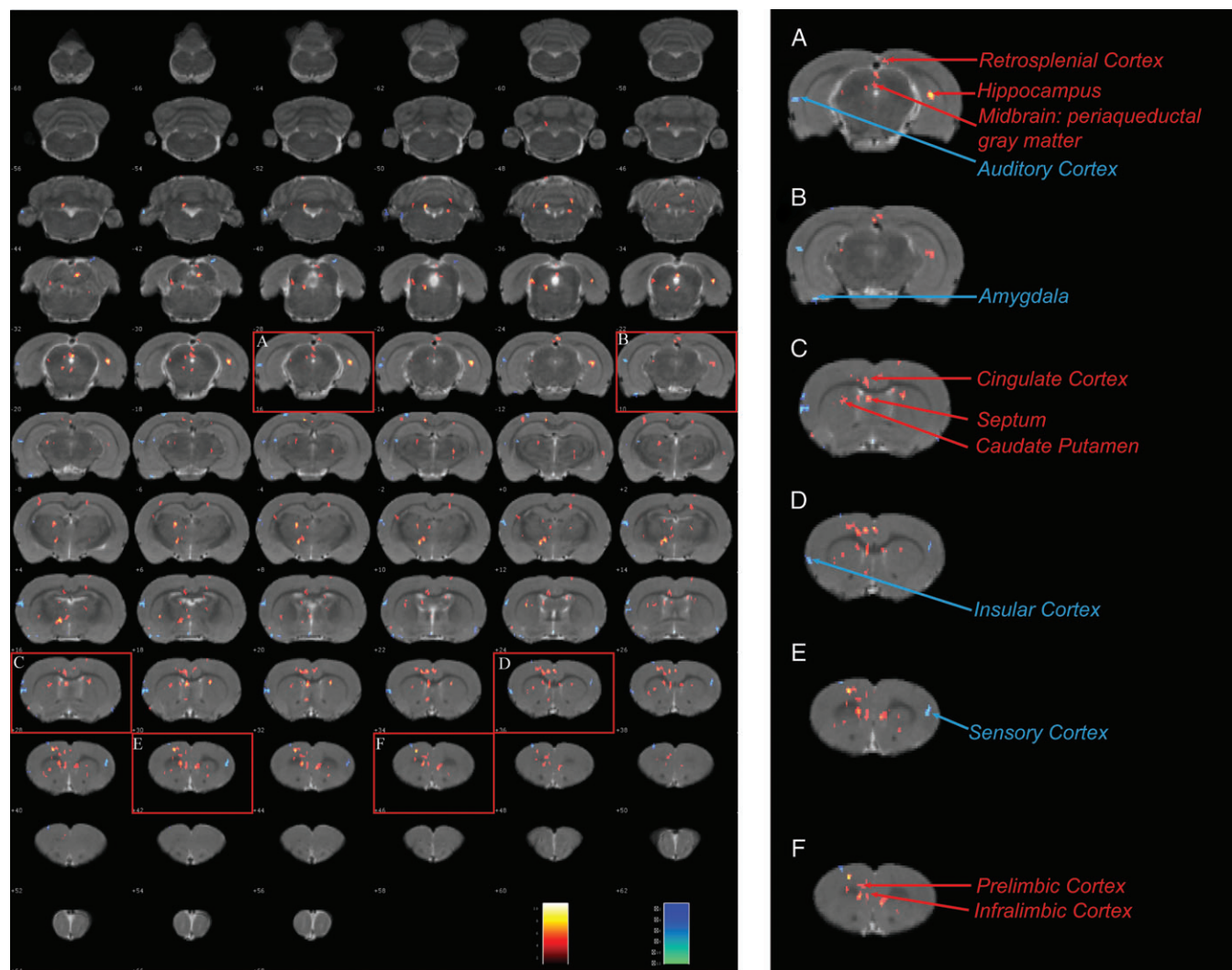


Fig. 2 Coronal view of the altered ReHo in the CUMS group compared to the control group. The voxel-level height threshold was $P < 0.005$ (uncorrected) and the cluster-extent threshold was 20 voxels. ReHo, regional homogeneity; CUMS, chronic unpredictable mild stress.

formulas. All drugs were administered by gavage once a day for 6 weeks. MXYS (10 g) was dissolved in 250 mL distilled water to a final concentration of 0.04 g/mL. The MXYS group was treated with MXYS (0.4 g/kg body weight), and the control and CUMS groups were treated with the same dose of saline.

Behavioral tests

At the end of the sixth week, all animals were subjected to the behavioral tests, performed and scored by three highly trained and experienced observers. The sucrose preference test (SPT), tail suspension test (TST) and forced swimming test (FST) were conducted as previously reported.⁶

fMRI data acquisition

As previously described,^{6,14} the mice were anesthetized using isoflurane in two stages: the induction stage (ratio of 1% oxygen + 3% isoflurane) and the continuous anesthesia stage (0.5% oxygen + 2.5% isoflurane). Afterward, mice were fixed to minimize head motion using a custom-made MRI-compatible cradle with ear and bite bars. A pressure sensor (Model 1030 Monitoring & Gating System; SAI Instruments, Stony Brook, NY, USA) was used to monitor the body temperature and respiration rates. The depth of anesthesia was also adjusted to maintain a consistent level according to the respiration rate. fMRI was performed on a 7-T Animal MRI Scanner (70/16 PharmaScan; Bruker Biospin GmbH, Rheinstetten, Germany), configured with a 38 mm birdcage rodent brain

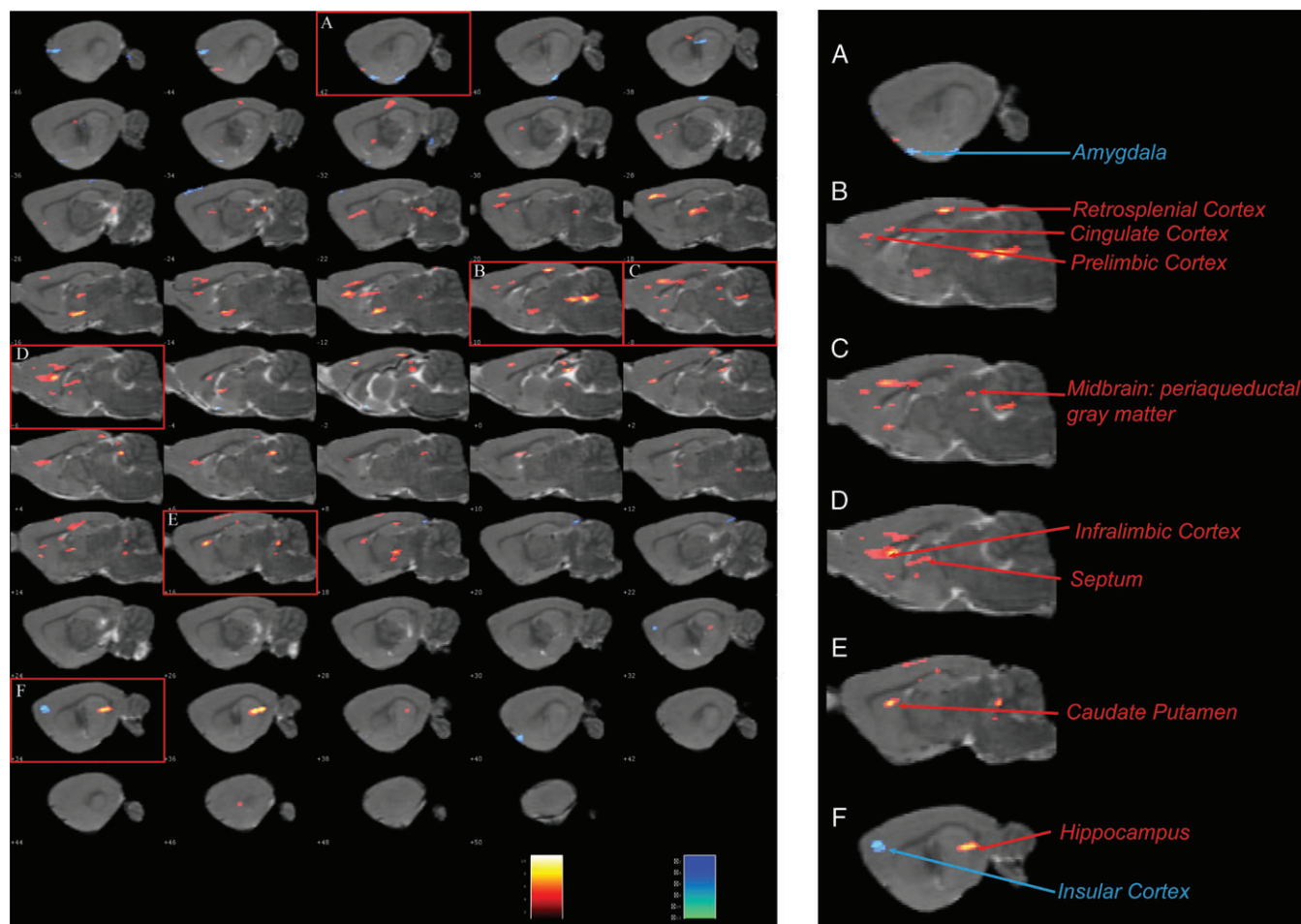


Fig. 3 Sagittal view of the altered ReHo in the CUMS group compared to the control group. The voxel-level height threshold was $P < 0.005$ (uncorrected) and the cluster-extent threshold was 20 voxels. ReHo, regional homogeneity; CUMS, chronic unpredictable mild stress.

quadrature resonator for radiofrequency transmission and reception. Echo-planar imaging with a low spatial and high temporal resolution was used for phase-image acquisition. Parameters were set as in previous studies:^{6,14} protocol = ax – T1w, resolution = $0.14 \times 0.14 \times 1.0$ mm, matrix size = 192×128 , slice thickness = 1.40 mm, slice gap = 0.10 mm, repetition time = 603.94 ms, echo time = 9.01 ms, averages = 32, scan time = 5 min 10 s, repetitions = 1, and volume = 1. A statistical parametric mapping toolbox (plug-in), the SpmratIHEP toolkit, was used to preprocess the phase images.^{19–21}

Statistical analysis

All statistical analyses were performed using the SPSS 20.0 software (IBM Corp., Armonk, NY, USA). Data are presented as means $\pm$ standard error of the mean (SEM). For the fMRI analysis, ReHo images were acquired across different bands. Statistical significance was set at an

uncorrected voxel-level height threshold of $P < 0.005$, and 20-voxel threshold was set for the cluster extent.

RESULTS

Modified Xiaoyaosan ameliorates CUMS-induced depressive-like behaviors

The CUMS model has been shown to induce several behavioral and physiological depressive-like symptoms. In our research, CUMS-treated mice showed a significant decrease in body weight compared to the control group, and MXYS administration significantly improved body weight against CUMS treatment (Fig. 1A). Our study revealed that the CUMS mice exhibited decreased sucrose consumption. Furthermore, the immobility time was increased both in TST and forced swim test in mice receiving the program. These results suggest that CUMS solution induce depressive-like behaviors in mice (Fig. 1B–D). MXYS enhanced the percentage of sucrose consumption as compared to the CUMS group (Fig. 1B). MXYS

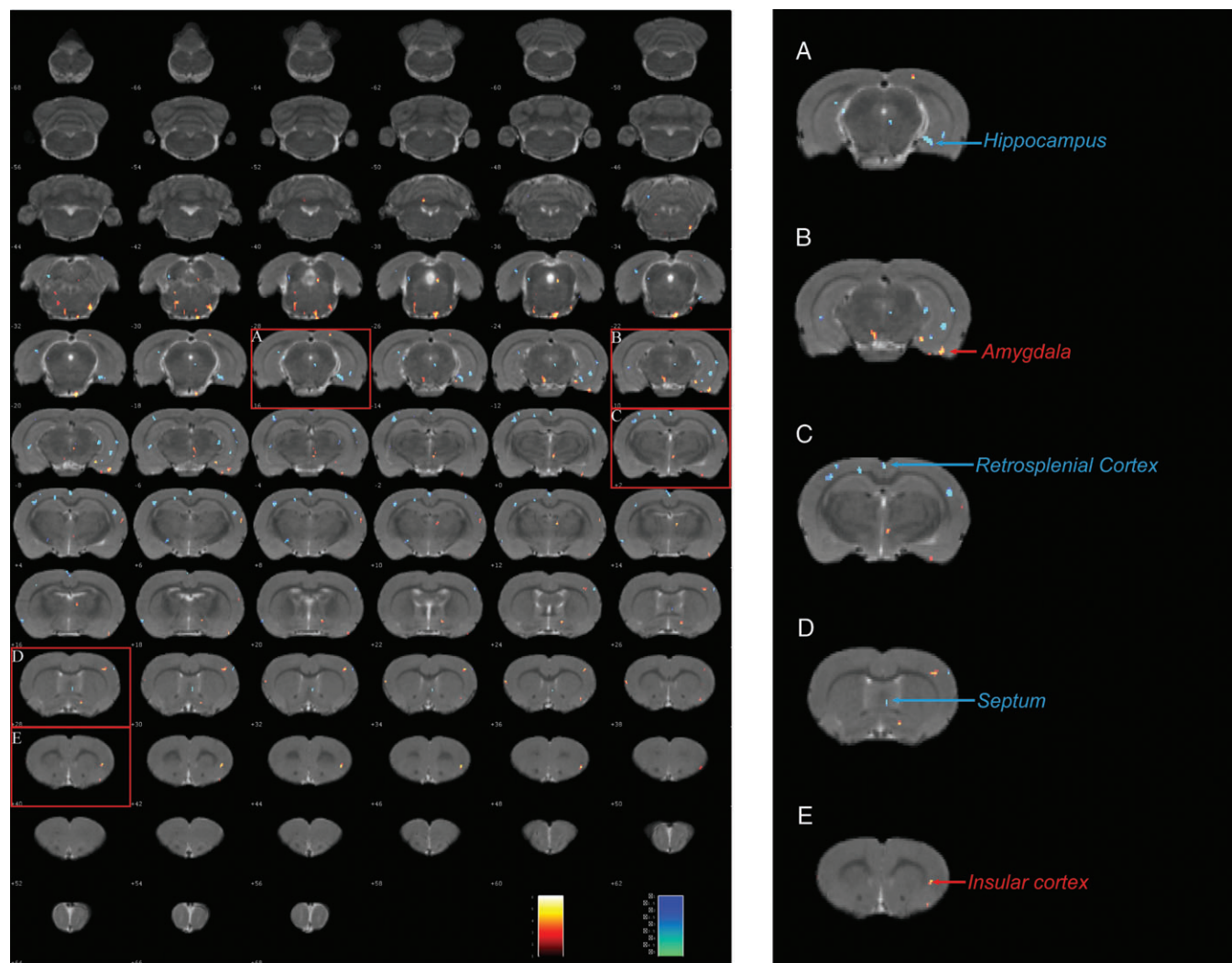


Fig. 4 Coronal view of the altered ReHo in the CUMS + MXYS group compared to the CUMS group. The voxel-level height threshold was $P < 0.005$ (uncorrected) and the cluster-extent threshold was 20 voxels. ReHo, regional homogeneity; MXYS, modified Xiaoyaosan; CUMS, chronic unpredictable mild stress.

decreased the immobility time of TST and FST (Fig. 1C, D). Collectively, MXYS can significantly ameliorate CUMS-induced anhedonia, loss of interest, psychomotor retardation and behavioral despair.

Disrupted ReHo in the coronal view of CUMS mouse brains

Besides the depression-like behaviors previously described,⁶ widespread ReHo disruptions were seen in several brain regions of CUMS mice. Compared to the control group, ReHo analysis in the coronal view of the CUMS mouse brains uncovered increased ReHo primarily in the limbic system and midbrain nuclei, including the infralimbic, prelimbic, cingulate, and retrosplenial cortices; hippocampus, septum, caudate putamen, and periaqueductal gray (PAG) matter (Table 2; Fig. 2). Other

brain regions with less significant ReHo increase are listed in Table 2.

In contrast, decreased ReHo was found primarily in a more limited range, including the sensory, insular, and auditory cortices and the amygdala (Fig. 2).

Disrupted ReHo in the sagittal view of CUMS mouse brains

We also performed ReHo analysis in the sagittal view of the CUMS mouse brains (Fig. 3) and replicated all the dysfunctional results mentioned in the coronal view. In addition, the sagittal view also revealed some other brain regions with significant ReHo disruptions, which are neglected in the ReHo analysis of the coronal view. Such brain regions included decreased ReHo in the flocculonodular lobe of the cerebellum (Fig. 3).

MXYS partially reversed aberrant ReHo in the coronal view of CUMS mouse brains

Compared to the widespread ReHo alterations in the brains of CUMS mice, ReHo analysis in the coronal view revealed that MXYS merely rescued aberrant ReHo in much more restricted limbic regions (Fig. 4). MXYS mainly decreased ReHo in the retrosplenial cortex, hippocampus, and septum and primarily increased ReHo in the hypothalamus, basilar part of the pontine, amygdala, and insular cortex (Fig. 4).

MXYS partially reversed aberrant ReHo in the sagittal view of CUMS mouse brains

We also performed ReHo analysis in the sagittal view of CUMS mouse brains to confirm the discoveries in the coronal view. In addition to reliably reproducing the ReHo disruptions, ReHo analysis in the sagittal view also uncovered several other brain regions, including increased ReHo in the superior colliculus and olfactory cortex and

decreased ReHo in the subiculum and visual and parietal cortices (Fig. 5). Other brain regions with less significant ReHo alterations are listed in Table 3.

DISCUSSION

In this study, we employed the recently developed ReHo analysis to evaluate the aberrant intra-region activity in the brains of CUMS mice. Despite the abundant evidence on disrupted ReHo in patients with depression,^{22,23} reliable animal models of depression capable of mimicking these ReHo alterations in humans are lacking. The decreased ReHo in the insular cortex and cerebellum supports the findings of previous reports with patients with depression,^{24–27} indicating that CUMS may reliably replicate the abnormal ReHo seen in the brains of patients with depression. Notably, the decreased ReHo in the amygdala of CUMS mice suggests deficits within the reward-related neural circuits centered at the amygdala, which were recently discovered and complement the

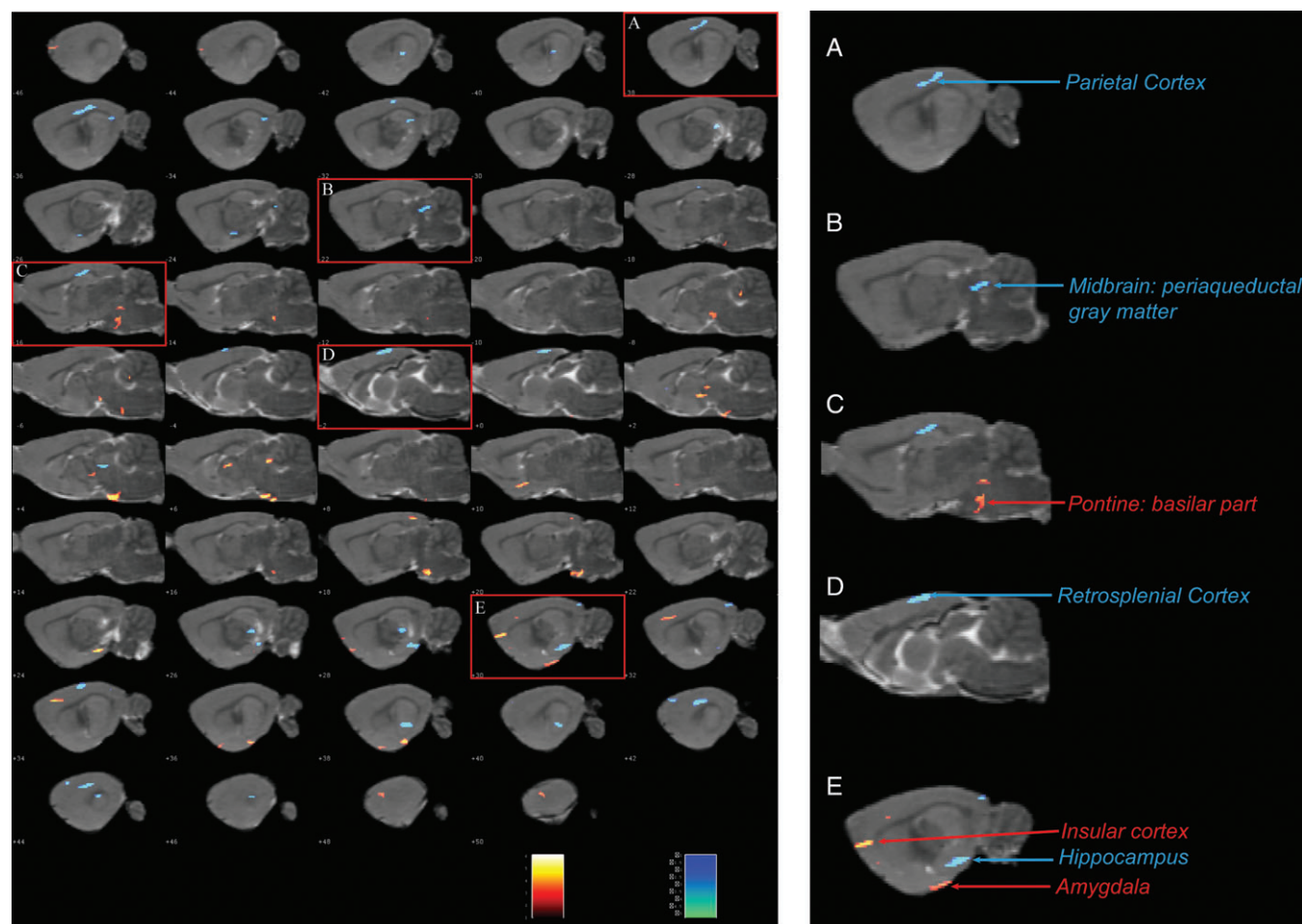


Fig. 5 Sagittal view of the altered ReHo in the CUMS + MXYS group compared to the CUMS group. The voxel-level height threshold was $P < 0.005$ (uncorrected) and the cluster-extent threshold was 20 voxels. ReHo, regional homogeneity; MXYS, modified Xiaoyaosan; CUMS, chronic unpredictable mild stress.

Table 3 Quantitative analysis of ReHo between CUMS + MXYS and CUMS groups

Group	Increased structural name	T-average	Coordinates of the peak voxel			Decreased structural name	T-average	Coordinates of the peak voxel		
			X	Y	Z			X	Y	Z
MXYS vs. CUMS	Hypothalamus	3.4646	-0.63176	6.6201	-3.4779	Subiculum	5.0759	3.5574	7.0118	-6.8379
	Midbrain: superior colliculus	4.5128	2.2077	5.243	-5.8779	Commissural stria terminalis	5.7021	4.7139	5.3228	-3.4779
	Olfactory cortex	8.0546	-5.7811	5.2633	-8.0379	Hippocampus	7.8066	3.4072	2.9549	-5.6379
	Pontine: basilar part	7.0559	0.90761	10.0729	-8.5179	Midbrain: periaqueductal gray matter	3.4312	0.61392	5.0682	-6.5979
	Insular cortex	5.1278	3.9566	4.7091	3.0021	Orbital cortex	3.1543	1.6748	3.9298	3.7221
	Amygdala	3.4136	4.186	9.2772	-3.9579	Parietal cortex	11.0426	-2.2322	0.7077	-3.7179
						Retrosplenial cortex	6.3713	-0.24073	0.3003	-2.7579
						Sensory cortex	7.1482	6.0108	4.9985	-0.5979
						Visual cortex	8.0204	-2.2256	0.60428	-4.1979

The voxel-level height threshold was $P < 0.005$ (uncorrected) and the cluster-extent threshold was 20 voxels. CUMS, chronic unpredictable mild stress; MXYS, modified Xiaoyaosan; ReHo, regional homogeneity.

traditional roles of the amygdala in encoding fear and aversion.²⁸ In addition, the finding of brain regions with increased ReHo in CUMS mice is consistent with the results of previous studies on dysfunctional connectivity in the depressive state,²⁹ including in the hippocampus and several cortical areas. Intriguingly, increased ReHo in the PAG might reflect increased social avoidance under the depressive state, since the PAG encodes behavioral avoidance.³⁰

Furthermore, the additional aberrant brain regions discovered in the ReHo analysis of the sagittal view suggest the necessity for an intact analyzing assay.

Importantly, MXYS only reversed ReHo limitation in several brain limbic regions, which highlights the essential roles of these regions in mediating the antidepressant effects of MXYS. Notably, the decreased ReHo in the retrosplenial cortex, hippocampus, and septum and the increased ReHo in the amygdala and insular cortex are exactly the opposite of CUMS-induced alterations. Such discovery indicates that MXYS might normalize the disrupted ReHo in these brain regions to exert antidepressant effects. Intriguingly, altered ReHo in the hypothalamus and basilar part of the pontine were not found in CUMS mice, whereas these two brain regions exhibited increased ReHo under MXYS treatment. Further investigations in other animal models of depression are required to replicate our discoveries, which is also essential for the precise construction of ReHo disruptions under the depressed state.

In summary, the present study provides a framework for analyzing ReHo in animal models of depression and uncovered aberrant connectivity, rescued by MXYS, in several brain regions. Our discovery presents a new perspective for the investigation of the antidepressant effects of traditional Chinese medicinal herbs.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (81873170 and 81230085), and the Scientific Research Initiative Program of Southern Medical University (PY2016N001), and Department of Education of Guangdong Province-Project for young innovation talents (2017KQNCX043).

AUTHORS' CONTRIBUTIONS

Zhiping Lv and Lei Gao participated in conception and design of the study; Yanmeng Bi, Zhaoyang Dong and Tingting Gao participated in generation, collection, assembly, and interpretation of data; Peng Huang participated in drafting and revision of the manuscript; Zhaoyang Dong, Tingting Gao, and Shaohui Huang participated in statistical analysis; Yanmeng Bi, Peng Huang and

Zhaoyang Dong contributed equally to this work and are co-first authors.

DISCLOSURE

The authors declare no conflicts of interest.

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